Table 2.2: Parameters for the outer solar atmosphere. Data from Minnaert (1953)

$\overline{\tau}$	T [K]	$\log P_{\rm gas}$	$\log P_e$	ρ [g cm^{-3}]	d [km]
0.01	4650	3.74	−0.15	1.4 10^{-8}	−412
0.02	4700	4.01	0.09	2.7	−325
0.06	4790	4.30	0.36	5.0	−230
0.10	4890	4.43	0.50	6.6	−187
0.20	5090	4.60	0.71	9.4	−129
0.32	5290	4.71	0.87	11.7	−89
0.40	5400	4.77	0.96	13.1	−67
0.60	5660	4.86	1.15	15.4	−32
0.80	5870	4.91	1.32	16.6	−12
1.00	6070	4.94	1.48	17.3	0

d = depth considered from the outside, with $d = 0$ at $\tau = 1$.
$P_e = n_e kT$ and can be derived from spectral lines (see Ch. 3).

2.8 Opacity and the absorption coefficients

Absorption is an expression used for the sum of true absorption (κ_ν) and scattering (σ_ν). The dominant form is in most stellar gases the true absorption, and is presented first. There are several forms of absorption: absorption due to ionization of atoms, absorption due to electronic transitions in atoms and ions (see Ch. 3), absorption due to dissociation of molecules, and absorption in free-free transitions. A special case is the absorption due to ionization of H^- (see below).

2.8.1 Absorption due to ionization

Atoms can be ionized by photons having an energy larger than the ionization energy $E_{\rm ion}$ of the actual excitation state of the atom. Excess energy will be transferred to the ion and the electron as kinetic energy $E_{\rm ph} = h\nu = E_{\rm ion} + \frac{1}{2}m_e v^2 + \frac{1}{2}m_{\rm ion}v^2$. The kinetic energy becomes part of the thermal energy of the gas. The process is also described as the bound-free transition (b-f). The reverse process is the recombination (free-bound, f-b), in which photons are produced.

Ionization can take place for frequencies $\nu > \nu_{\rm ion} = E_{\rm ion}/h$ and this produces a sharp depression, the **ionization edge**, in the spectral continuum. The absorption coefficient of hydrogen is (Kramers in 1923)

$$a_n = \frac{1}{n^2}\frac{64\pi^4}{3\sqrt{3}} \cdot \frac{1}{\nu^3} \cdot \frac{Z^4 m_e e^{10}}{ch^6 n^3}\ g \tag{2.40}$$

with g the Gaunt factor (a factor with which deviations from a simple formula are taken care of). Clearly, the absorption coefficient is proportional to ν^{-3} so that, starting at the ionization edge, the absorption due to ionization becomes smaller with larger frequency.

2.8.1.1 Total absorption cross section for hydrogen

Since atoms may be in an excited state, ionization can take place also from this excited state. This leads to another ionization edge at that limit. The frequency of the ionization edges of hydrogen-like atoms (H, He, C^{+5}, etc.) can be given by

$$\nu_{\rm edge} = R\ Z^2\ \frac{1}{n^2} \tag{2.41}$$

in which R is the Rydberg constant and Z the nuclear charge. For $n = 1,\ 2,\ 3 \ldots$ the edges are called after Lyman, Balmer, ... (see Fig. 2.5).

For the interpretation of observations of the shape of the absorption edges of hydrogen see, e.g., Ch. 3.3.2. There it is also shown that the level of hydrogen ionization in the stellar atmosphere

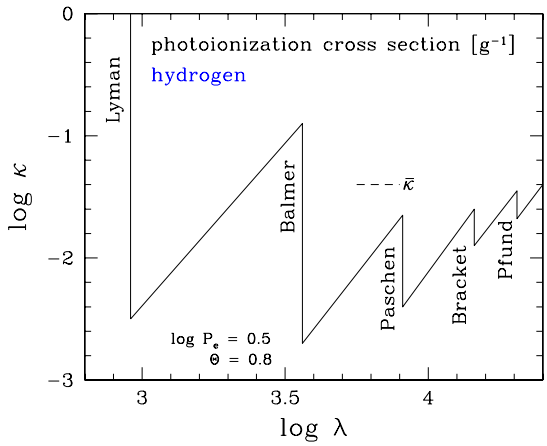

Figure 2.5: The absorption coefficient of hydrogen is plotted in logarithmic scale ($\log \kappa$ [g^{-1}] vs. $\log \lambda$). The absorption edges from the levels $n = 2$ (Balmer), $n = 3$ (Paschen), $n = 4$ (Brackett), and $n = 5$ (Pfund) are marked. The model used has $\Theta = 0.8$ ($T_{\rm eff} \simeq 6000$ K) and $\log P_e = 0.5$. The Rosseland mean opacity, $\overline{\kappa}$ (see Sect. 2.6.4), is indicated.

plays a role only in a certain temperature range: hydrogen is fully ionized for $T \geq 20\,000$ K and the ionization edges have disappeared, while for $T \leq 6000$ K hydrogen is not even excited to level $n = 2$ so that the Balmer and higher n absorption edges are not present.

2.8.1.2 Absorption due to ionization of helium

The ionization energy of neutral He is 25.4 eV ($\lambda = 506$ Å). In the solar atmosphere the level of excitation of He is vanishingly small, so that He absorption plays no role in the atmosphere itself.

In hot stellar atmospheres He is quite visible. There are twice as many He ionization edges as H edges, and every second edge coincides with an H edge (H and He have nearly equal Rydberg constants). This is immediately clear using Eq. 2.41:

$$\text{He II}: \ \nu \simeq \frac{4}{n^2} \quad \text{and} \quad \text{H}: \ \nu \simeq \frac{1}{n^2} \tag{2.42}$$

He has edges at 228, 911, 2050, 3644 and 5694 Å. The 3644 Å edge is called the Pickering edge. These edges have consequences for the overall opacity in stellar gas (see Fig. 2.8 and Ch. 4.4.1).

2.8.1.3 Absorption due to ionization of metals

With the metals, the more important ones are those of abundant elements. As an example, consider the atmosphere of the Sun, in which the metals are neutral or singly ionized. Fe I has an I.P.= 7.90 eV so the ionization edge is at $\lambda = 1570$ Å. Excitation to higher levels is easy so that ionization can take place at almost any wavelength. Si I has an I.P. = 8.15 eV thus an edge at $\lambda = 1520$ Å. Mg I has I.P. of 7.65 eV, edge at $\lambda = 1630$ Å, and a clear edge from an excited level at $\lambda = 2520$ Å. In hot atmospheres similar considerations can be made for the higher ionic states.

Ionization potentials are listed in "Allen's Astrophysical Quantities" (Cox 2002). Information about further energy levels is available in Moore (1959) or Wiese et al. (1966, 1969).

Note that the metals impose numerous absorption lines onto the spectrum. In many cases, there are so many that individual lines hardly can be recognized and the absorption looks like continuous absorption. An example is given in Fig. 2.6.

2.8.2 The H$^-$ ion

In gas near 5000 to 6000 K most of the metals are singly ionized. This leads to some electron density. In such gas, depending on density and availability of electrons, hydrogen may build the 'radical' H$^-$, the H$^-$ ion. The binding energy of H$^-$ is 0.75 eV and there are no excited states of H$^-$ below that energy. One can apply the Saha equation (Ch. 3.2.2) to calculate the number of H$^-$ ions in the atmospheres of stars. For the Sun one so arrives at $n(\text{H}^-) \simeq 3 \cdot 10^{-8}\ n(\text{H})$. For details see, e.g., Böhm-Vitense (1989, Ch. 7).

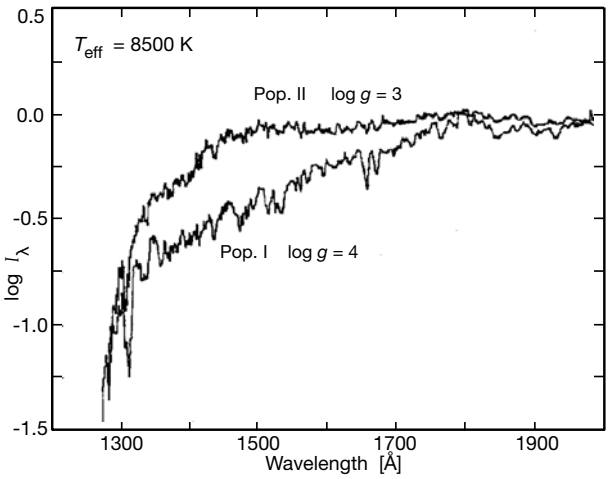

Figure 2.6: The absorption coefficient κ due to metals is mostly due to individual spectral lines. There are, however, so many that the effect of their overlapping is like that of continuum absorption. The average spectral energy distribution in the UV of stars of Population I is shown together with that of stars of Population II, the latter being metal poor by about a factor 10, thus [M/H]$\simeq -1$. The UV intensity of the metal poor stars is higher because of substantially smaller overall opacity. The spectra were normalised in the V-band. Figure from Huenemoerder et al. (1984).

This ion is, of course, easily dissociated which requires photons with an energy of more than 0.75 eV. This radiative dissociation produces a very broadly spread contribution (centered on $\lambda \simeq 16\,500$ Å) to the opacity of the atmospheric gases (see Fig. 2.8, left panel). It leads to a broad and shallow depression in the spectral brightness of the Sun in the near IR part of the spectrum.

2.8.3 Absorption due to dissociation

Molecules can dissociate through absorption by photons having an energy larger than the dissociation energy, $h\nu > E_{\rm diss}$. This absorption takes place over a large range of frequencies and leads therefore also to continuum absorption. If molecule AB is dissociated into the atoms A and B by a photon $h\nu$, the energy of the photon is not only used for the dissociation ($AB + h\nu_{\rm diss} \rightarrow A + B$) but the excess energy is taken up as kinetic energy or possibly as energy to excitate the atoms remaining: $E_{\rm ph} = E_{\rm diss} + E_{\rm kin} + E_{\rm exc}$.

Dissociations have a certain probability to take place. This is characterized by a dissociation constant K as

$$K_{AB} = \frac{P_A P_B}{P_{AB}} = kT \frac{n_A\ n_B}{n_{AB}} \quad . \tag{2.43}$$

the latter part of the equation assuming $PV = nkT$. The abundance of various molecules in cool stellar atmospheres is shown in Fig. 2.7, some dissociation constants are listed there, too. Examples of spectral line absorption by molecules can be found in Figs. 3.10 and 10.21.

2.8.4 Free-free transitions

At higher temperatures and higher electron densities, electrons passing by ions are accelerated in the Coulomb field and then radiate the *Coulomb-Bremsstrahlung*. This process is also indicated as a free-free transition (f-f), producing free-free radiation. Similarly, energy can be absorbed from the photon field, leading to acceleration. The absorption coefficient for this process, in the case of fully ionized gas (in stellar interiors) and with 80% H and 20% He by mass, is

$$a_{\rm ff} = 1.32 \cdot 10^{-2} \frac{n_e^2}{T^{3/2}} \frac{1}{\nu^2} g \tag{2.44}$$

again with g as a correction factor ($g \simeq 1$).

2.8.5 Scattering

Some of the photons are scattered by the gas particles. There are several kinds of scattering.

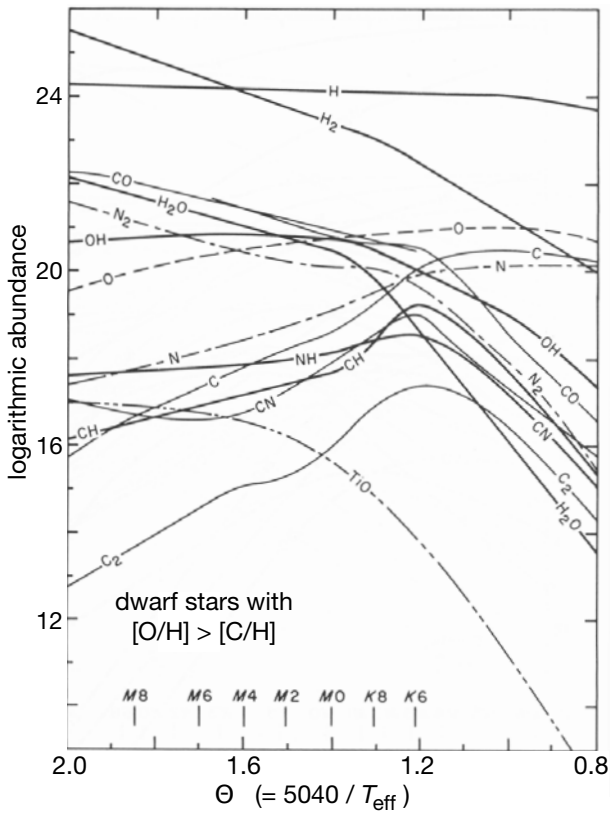

Figure 2.7: The abundance of molecules in cool stellar atmospheres is indicated in a logarithmic scale, along with the abundance of the relevant elements (for an atmosphere richer in O than in C). The temperature is given as Θ (see Ch. 3.2.2) so that the temperature range shown is from $T_{\rm eff} \simeq 2500$ K at the left to $T_{\rm eff} \simeq 6300$ K at the right. Figure from Aller (1963).

Molecules may be dissociated, requiring a certain energy depending on the excitation condition of the gas (see Eq. 2.43). Dissociation constants for two molecules are given below.

Dissociation constants for H_2 and TiO (as given by Aller 1963)			
T [K]	$\log kT$	$\log K$	
		H_2	TiO
5000	−0.37	7.7	. .
4000	−0.40	6.4	3.7
3000	−0.59	4.4	0.7
2000	−0.76	0.3	−5.3

Photons may have 'resonant' scattering on atoms and ions. This process is rather absorption and instantaneous re-emission of the photon. It takes place at frequencies around the frequency of an electronic transition (an absorption line). The cross section is

$$\sigma_R(\nu) = \frac{8\pi\ e^4}{3m_e^2c^4}\left(\frac{\nu}{\nu_0}\right)^4 N \tag{2.45}$$

with ν_o the resonance frequency of a transition (e.g. 1215 Å for the hydrogen Lyman-α line).

Photons may be scattered by free electrons (*Thomson scattering*) or by a molecule (*Rayleigh scattering*). The cross section is

$$\sigma_e = \frac{8\pi\ e^4}{3m_e^2c^4}\ n_e = 6.65 \cdot 10^{-25}\ n_e \tag{2.46}$$

with n_e the electron density. Since in the atmosphere of the Sun the relative number of electrons is small ($n_e/n_{\rm atom\ +\ ion} \simeq 3 \cdot 10^{-3}$) there is hardly any Thomson scattering. In hot atmospheres hydrogen and metals are ionized so $n_e/n_{\rm nuclei} \simeq 1$ and Thomson scattering is important indeed.

Another scattering process is the *Compton scattering* in which a photon scatters on an relativistic electron and therby gains energy. This may be of relevance in the deep interior of stars.

2.8.6 Total absorption coefficient

The total absorption coefficient of gas in the broader visual domain of spectral energy distributions is presented in Fig. 2.8. Two cases are shown: the cool case ($\Theta = 1$, $T_{\rm eff} = 5040$ K) and the hot case ($T_{\rm eff} = 28\,300$ K).

For opacity at higher temperatures (rather pertaining to layers below the atmosphere and the conditions of the stellar interior) see Ch. 4.4.1 and Fig. 4.4.

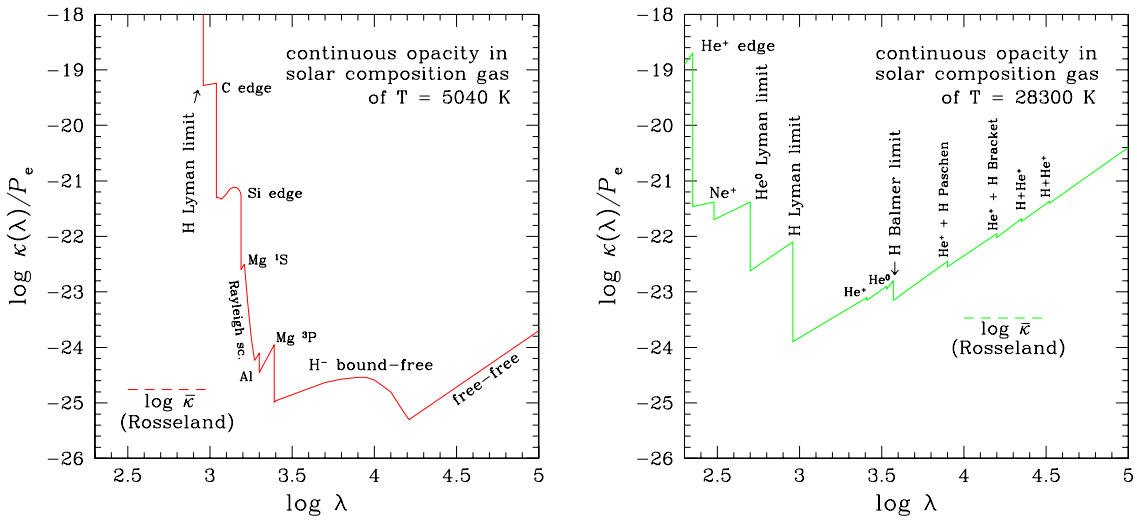

Figure 2.8: The total absorption coefficient κ/P_e [g^{-1}] due to continuous processes (foremost ionization). *Left*: gas at $T_{\rm eff} = 5040$ K. *Right*: gas at $T_{\rm eff} = 28\,300$ K. Various ionization edges are explained. Note that the Rosseland mean $\overline{\kappa}$ has different values for different temperatures. The figures are an adaptation of those of Unsöld (1977) who used diagrams from Vitense (1951) for a range of T as input; for sources of metal photoionization cross sections see, e.g., de Boer et al. (1972).

2.8.7 Effects of gas density on opacity

Most of the processes presented have a dependence on gas density. This may be, e.g., due to excitation of the state from which absorption takes place or due to a density dependence of the ionization or dissociation balance (for details on these mechanisms see Ch. 3.2). The opacity thus depends, for a given temperature, also on gas density. The effect is rather indirect and shows up clearly only under certain conditions.

One effect of gas pressure on the opacity is given in Fig. 2.9.

2.9 Emission and the emission coefficient

Radiation set free by hot gas can be described by the Planck function, because the gas is in LTE. Thus inside stars $j_\nu = B_\nu(T)$.

Further sources of emission are:

∘ free-free transitions or Coulomb-Bremsstrahlung (Ch. 2.8.4). Electrons passing by ions are being accelerated and thus emit radiation.

∘ free-bound transitions or recombination radiation. Each recombination produces a photon which, depending on the mean excitation state of such ions, is not easily absorbed.

Such further sources of emission become noticeable individually only when the conditions in the gas start to deviate from LTE (rather near the stellar surface). Remember that LTE is defined as the condition in which all radiation-matter interactions are in balance.

2.10 The spectral continuum and the Planck function

The source function, S_ν, of the gas in the interior of stars is the Planck function, because the gas is in LTE. However, in stellar surface layers the gas is not in LTE which results in an observable spectral energy distribution deviating considerably from the Planck like distribution emerging from the interior layers.

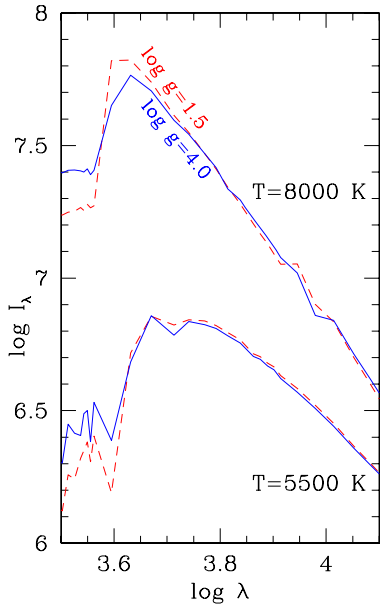

Figure 2.9: A comparison of the spectral energy distributions of cool main-sequence stars with those of red giants of the same temperature shows the effect of different gas density on the strength of the opacity due to H (the Balmer jump) as well as of the H^- ion. In the atmosphere of the red giant ($\log g = 1.5$) the lower gas density leads to less H^- and thus to larger n_e so to more H-atoms excited to the Balmer level. Thus the red giant has, in comparison, a larger Balmer jump (see Ch. 3.3.4 for the explanation of the Balmer level population). This explains the difference in location of the main sequence and the giant branch in an $U - B$ vs $B - V$ diagram. Graph using data from Kurucz (1992).

A simple method to imagine how the stellar surface spectral energy distribution looks like is to carry out a thought experiment. Starting with the Planck function one applies absorption of continuum nature like that given in Fig. 2.8. (Working with logarithmic representations one can just "substract" one curve from the other.) The absorption consists always of absorption by the metals, plus that by H and He depending on the temperature. For $T_{\rm eff} < 6000$ K there is the contribution by H^-. For stars with $T_{\rm eff}$ in the range of 6000 to 20 000 K the hydrogen atom contributes with absorption due to photo-ionization. In the visual part of the spectrum, the contribution of H leads around the Balmer jump (Fig. 2.5 and Ch. 3.3.4) to a temperature effect. Its effect on the spectral energy distribution can be seen clearly in the two-colour diagram, $U - B$ vs. $B - V$ (Fig. 2.10).

The next step is to add the effects of individual absorption lines (Ch. 3). Such are, in the middle temperature range, foremost the lines of the hydrogen Balmer series, but all other lines contribute, too. Especially when the density of absorption lines in the spectrum is large, the effect looks like that of a continuum depression, as in Fig. 2.6. Note that absorption lines are generally much stronger and much more abundant on the Wien side of the maximum than in the Rayleigh-Jeans part of the energy distribution.

2.10.1 Effects for the CMD

The Planck spectral energy distribution and its deviations show up also in the CMD (see Fig. 1.2 middle panel and Fig. 21.3). Going along the main sequence from the cool to the hot end one has the following effects.

Very cool end: When these stars are observed in B and V one measures rather at the short wavelength side of the intensity maximum. The spectral slope is steeply red, and the spectrum is loaded with absorption lines. Stars of slightly higher temperatures already have a considerably increased emitted intensity (in M_V).

Cool end: The B and V bands are still 'on the Wien side of the Planck function', but now $B - V$ changes quickly toward higher temperatures, while M_V changes less drastically. The main sequence levels off. Why does this differ from the behaviour of the very cool stars?

Middle temperature range: Stars are, in the visual, now near the peak of the intensity distribution. In this temperature range the photoionization of hydrogen dominates the structure of the atmospheric opacity causing the above described deviations of the Planck spectrum.

Hot end: The visual part of the spectral energy distribution is approaching that of the Rayleigh-Jeans approximation, and $B - V$ becomes a constant (Rayleigh-Jeans limit). For hotter stars M_V

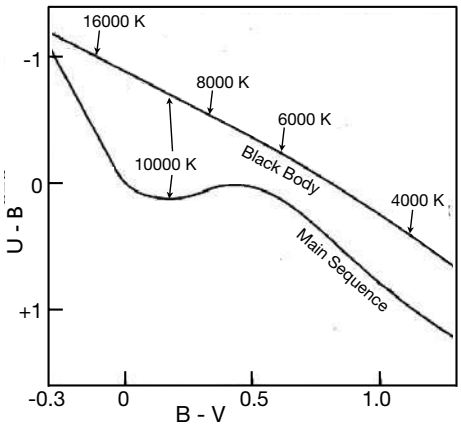

Figure 2.10: The $U-B$ vs. $B-V$ two-colour diagram shows how the colours of the Planck function change with temperature. The run of the colours of main-sequence stars deviates from the Planck colours, largely due to effects of opacity by the hydrogen ionization from the Balmer level (the Balmer jump). For details of the Balmer jump see Ch. 3.3.4. Note that when these effects play no role (toward the lower temperatures, especially $T < 4000$ K) the colour slope is the same as that for the Planck function. The locus of red giants in this diagram is different from that of main-sequence stars, because in their atmospheres density effects play a role for the opacity (see Fig. 2.9).

will be brighter still. This can be understood as being due to the larger stellar radius (more mass) as well as due to the *linear* dependence of M_V on T (Rayleigh-Jeans part of the spectrum!).

2.10.2 Backwarming, blanketing

One important aspect has to be added to the considerations above. When due to continuum opacity (and also spectral line opacity) energy is absorbed, the retained energy leads to an increase in temperature of those gases. This effect is in the literature called *backwarming* or *blanketing*. The higher temperature implies a higher temperature Planck function, so that all of the thought experiment has to be performed anew with that new spectral energy distribution. Clearly, in doing so, the luminosity must stay the same. Another way to phrase the effects is that due to opacity the emergent energy is redistributed over the spectrum. This is, in a way, the fullfillment of the continuity equation (Sect. 2.5). In general, the intensity taken away at the higher energy end of the spectral energy distribution is compensated for by an overall rise in the spectral energy distribution, which therefore is also noticeable in its lower energy part.

2.10.3 Electron density and opacity effects

Also the gas density plays a role in the size of the atmospheric opacity. At lower density, the fewer collisions between electrons and atoms or ions lead to a change in the ionization balance (see Ch. 3.2.2), e.g., in cooler atmospheres to less H in the form of H^-, or to less excitation (see Ch. 3.2). The combination of H^- and electron density is of interest here: with *fewer* H^- there are *more* free electrons thus more excitation of H to the Balmer level and so a larger Balmer jump. The effects for the spectral energy distribution in atmospheres of $T_{\rm eff} = 8000$ K and 5500 K can be seen in Fig. 2.9.

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